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[22TCSBS122]

FOURTH SEMESTER B.Tech EXAMINATION, APRIL - 2024

ENVIRONMENT AND ECOLOGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Define environment.	L-1	CO1	**02**
	b.	Why is biodiversity important?	L-1	CO2	**02**
	c.	Write any two social problems of dams.	L-1	CO3	**02**
	d.	What are volatile organic compounds? Mention any two sources of it	L-1	CO4	**02**
	e.	Describe scrubber. Write its importance.	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Explain the three classifications of biotic components.	L-1	CO1	**05**
		OR			
	b.	Describe the importance of environmental sciences.	L-2	CO1	**05**
3.	**a.**	Justify why India is one of the mega-diversity countries.	L-2	CO2	**05**
		OR			
	b.	Summarize the three levels of biodiversity.	L-2	CO2	**05**
4.	**a.**	What are the two types of natural resources? Write the differences between them.	L-2	CO3	**05**
		OR			
	b.	Illustrate the importance of alternative sources of energy.	L-2	CO3	**05**
5.	**a.**	Differentiate between ionizing and non-ionizing radiations.	L-1	CO4	**05**
		OR			
	b.	Explain the preventive measures for the radio-active pollution.	L-2	CO4	**05**
6.	**a.**	What are the positive impacts of a healthy environment on human health?	L-1	CO5	**05**
		OR			
	b.	Illustrate the concept of ozone layer depletion.	L-2	CO5	**05**
7.	**a.**	What are the key principles of sustainable development?	L-1	CO5	**05**
		OR			

	b.	What are the key functions of ecosystem?	L-1	CO1	05
		PART-C	**06 X 10=60**		
8.	a.	Describe the composition of atmosphere and lithosphere	L-2	CO1	10
		OR			
	b.	Write about the characteristics, features, structure and functions of grassland ecosystem and aquatic ecosystem.	L-2	CO1	10
9.	a.	Illustrate the strategies, advantages and disadvantages involved in in-situ conservation methods.	L-2	CO2	10
		OR			
	b.	Define bio-geographic zone. Discuss about productive use values and consumptive use values of biodiversity.	L-2	CO2	10
10.	a.	What are the benefits of dams? What are the reasons for water scarcity?	L-1	CO3	10
		OR			
	b.	Describe the key functions and uses of forest resources	L-2	CO3	10
11.	a.	Elaborate in detail about waste-water treatment with a neat diagram.	L-2	CO4	10
		OR			
	b.	Discuss different measures to control water and soil pollution.	L-2	CO4	10
12.	a.	Elaborate the disaster management and its various stages.	L-2	CO5	10
		OR			
	b.	Discuss about the acid rain with suitable equations and its impacts on environment.	L-2	CO5	10
13.	a.	Interpret the various threats to biodiversity.	L-2	CO2	10
		OR			
	b.	Discuss the source, effects and control techniques of noise pollution.	L-2	CO4	10